

ACTIVITY NO. 4

FACTORS AFFECTING THE RATE OF LOSS OF HEAT OF LIQUID

Aim: To study the factors affecting the rate of loss of heat of a liquid.

Apparatus: Two calorimeters of same area of cross-section, two thermometers, stand, hot water bath, liquid (water).

Newton's law of cooling :

The rate of loss of heat of a body is directly proportional to difference of temperature over the surrounding, provided the difference is small.

$$\frac{d\theta}{dt} \propto (\theta - \theta_0), \quad \frac{d\theta}{dt} = K (\theta - \theta_0)$$

Where K is constant, depend upon

i) mass of the liquid ii) specific heat of the liquid iii) temperature difference.

For same temperature difference, rate of cooling depends upon

- i) Area of the surface of the body.
- ii) Nature of the surface of the body.
- iii) Material of the body.

Procedure :

Perform experiment Newton's laws of cooling and repeat the experiment as per following condition (any one condition)

Condition :

Take same amount of liquid in similar calorimeters, one having black painted surface and other having polished surface. (Area and material same.)

Observation :

- Least count of thermometer = 0.2°C
- Least count of stop clock = 0.01 s

Ob. No.	Time in min. open	Temperature of water $^\circ\text{C}$	Ob. No.	Time in min. closed	Temperature of water $^\circ\text{C}$
1	0	80	9	0	66
2	1	75	10	1	66
3	2	71	11	2	65
4	3	69	12	3	64
5	4	67	13	4	63
6	5	64	14	5	62
7	6	62	15	6	61
8	7	60	16	7	60
17	8	57	24	8	59
18	9	55	25	9	58
19	10	54	26	10	57
20			27		
21			28		
22			29		
13			30		

Plot the graph of temperature against time on the same graph paper for liquids in two calorimeters. Compare the cooling curves.

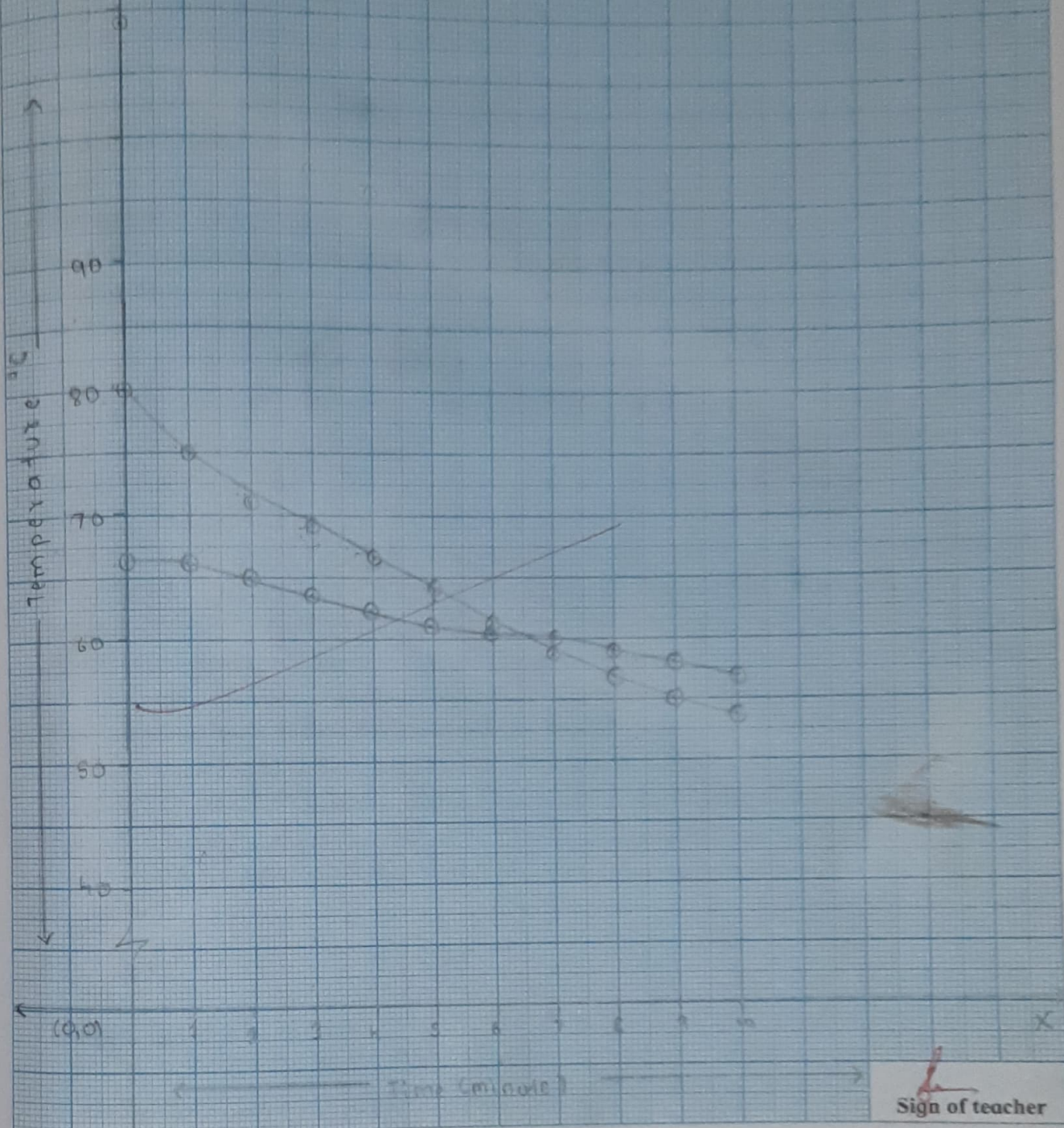
Scale :

X-axis

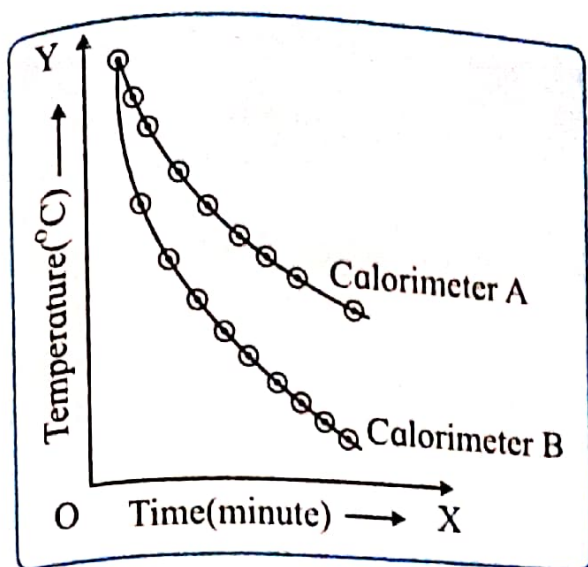
1 cm = 1 min

Y-axis

1 cm = 5°C



Sign of teacher



Scale :

On X-axis: 1 cm = 1 min.

On Y-axis: 1 cm = 5 °C

fig. A 3.1: Temperature and for graphs for calorimeter A, having small/ polished surface area and calorimeter B having large/ blackened surface area

Conclusion:

Cooling is faster in case of a calorimeter, painted black as compared to cooling in calorimeter whose surface is polished.

Precautions :

1. Make sure that the bulb of thermometer is well inside the water.
2. The enclosure should have proper insulation to avoid loss of heat due to conduction or convection from hot water.
3. The thimble or calorimeter should be dry from outside.

Questions

1. What do you meant by radiant heat?

Radiant heat is the heat energy transmitted from one body to another in the form of electromagnetic waves without requiring any material medium.

2. Why is cooling faster in the calorimeter painted black?

A black surface is a good emitter of heat radiation. Therefore, a calorimeter painted black loses heat more rapidly by radiation and cools faster.

Remark and sign of teacher :

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18/11/22 Excellent